RentSphere Formal UML Architecture Specifications

Project: RentSphere: Decentralized P2P Asset Rental & 8x8 Micro-Storage Engine
Institution: Goa University — Department of Computer Science & Technology
Program: Master of Computer Applications (MCA) — Object-Oriented Technology Lab (Assignment 3)
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Diagram Format: Mermaid.js (Standard Markdown Embedded)

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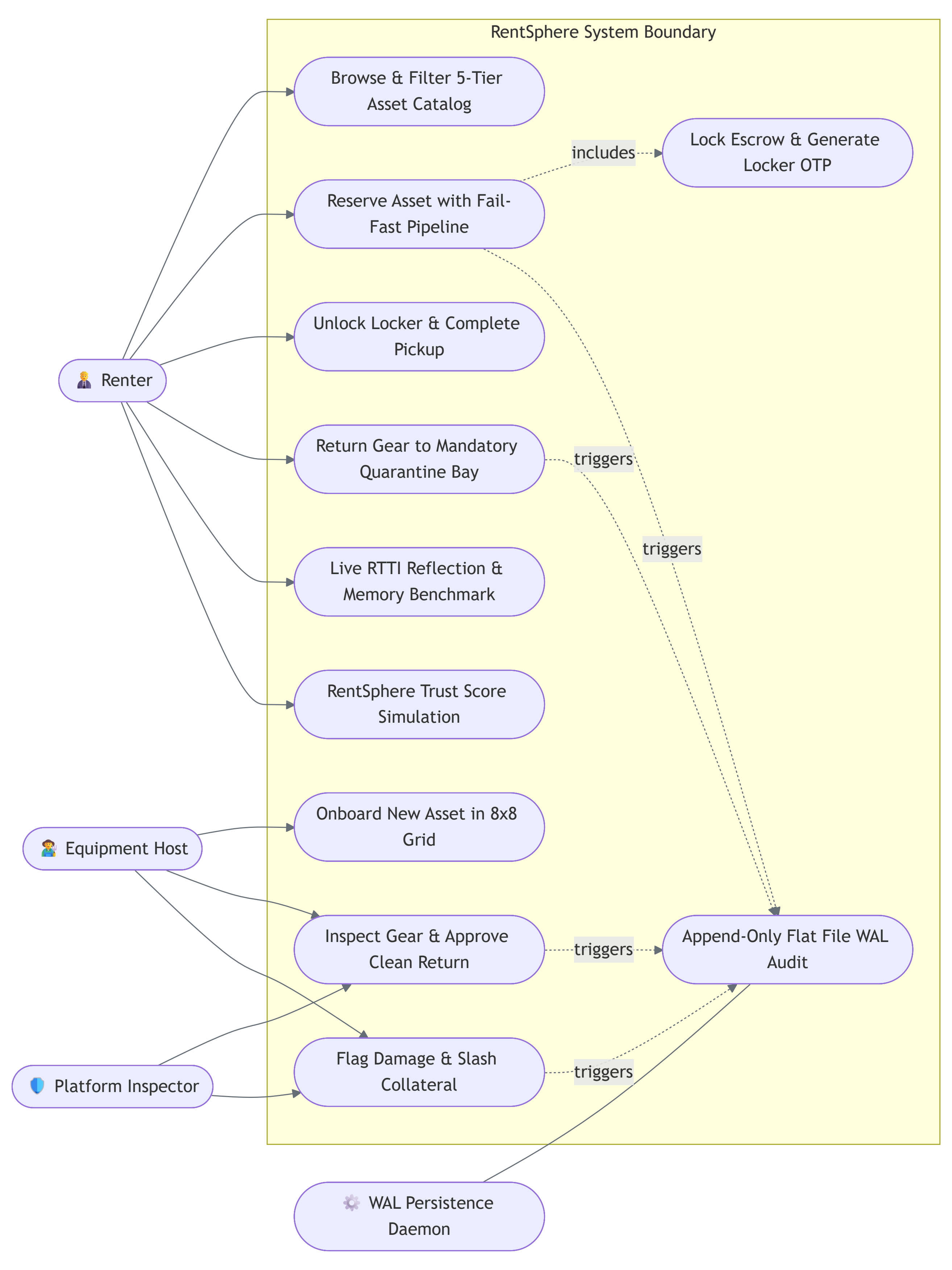
1. Use Case Diagram

Models platform actors (Renter, Equipment Host, Platform Inspector, WAL Engine) and their system interactions.


RentSphere System Boundary
Renter
Equipment Host
Platform Inspector
Browse & Filter 5-Tier Asset Catalog
Reserve Asset with Fail-Fast Pipeline
includes
Lock Escrow & Generate Locker OTP
Unlock Locker & Complete Pickup
Return Gear to Mandatory Quarantine Bay
triggers
triggers
Live RTTI Reflection & Memory Benchmark
RentSphere Trust Score Simulation
Onboard New Asset in 8x8 Grid
Inspect Gear & Approve Clean Return
triggers
Append-Only Flat File WAL Audit
Flag Damage & Slash Collateral
triggers
WAL Persistence Daemon

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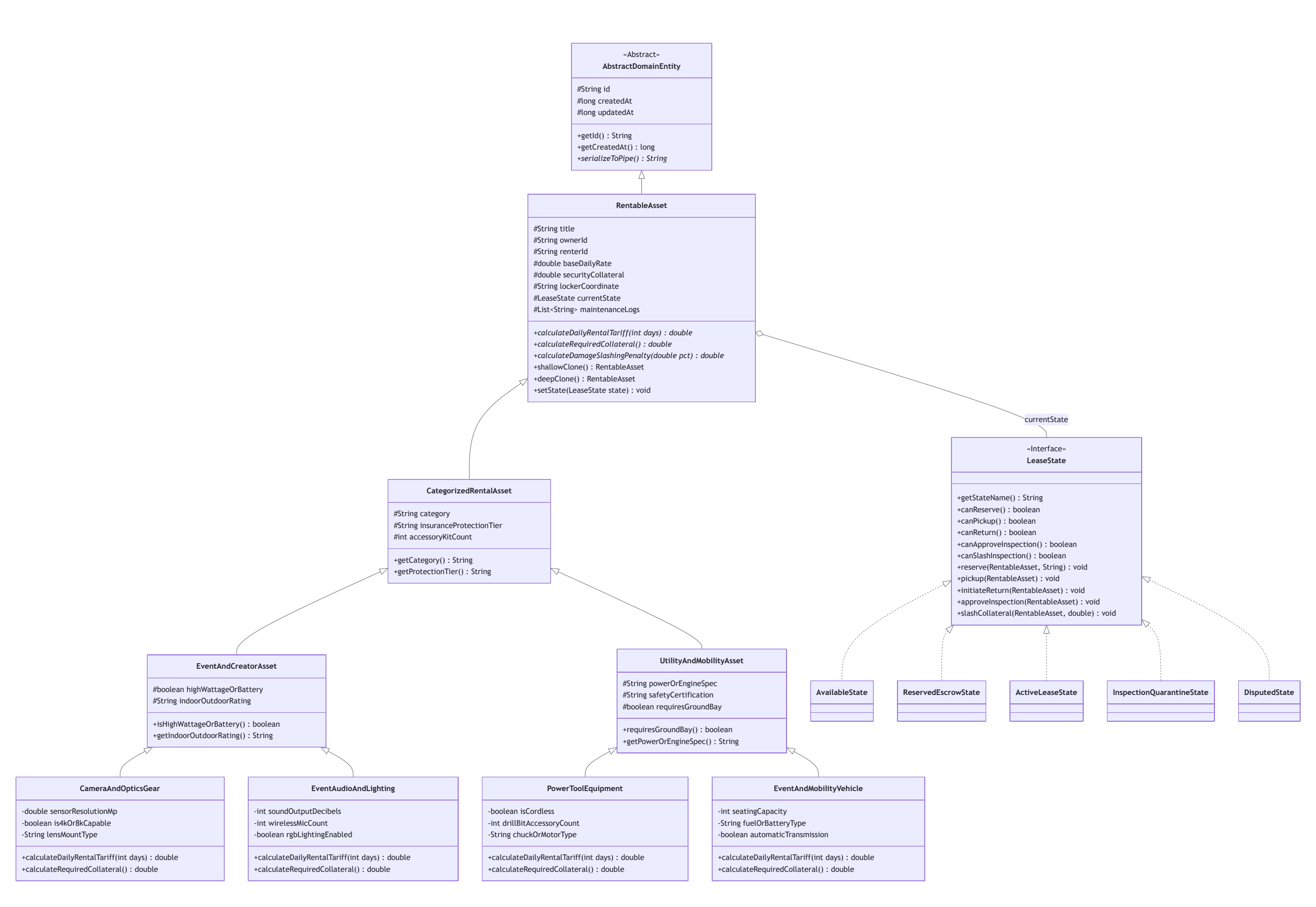
2. 5-Tier Object-Oriented Class Hierarchy Diagram

Models the complete 5-tier inheritance architecture, GoF state composition, and escrow/locker associations.



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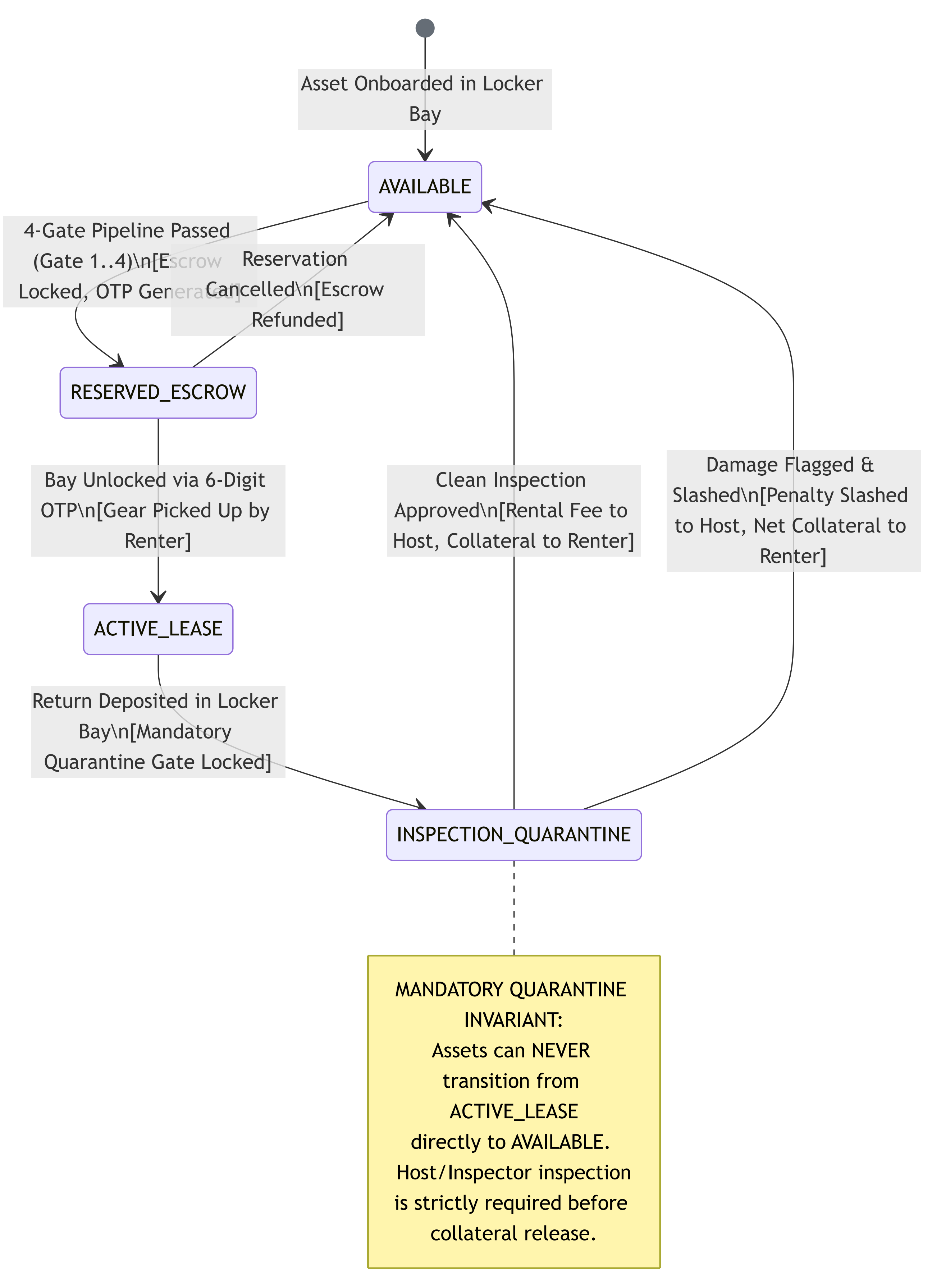
3. GoF 4-Stage Statechart Diagram (Mandatory Quarantine Invariant)

Models state transitions, showing that returning active gear **must always** transition to `INSPECTION_QUARANTINE` first. Returning directly to `AVAILABLE` is strictly forbidden.


Asset Onboarded in Locker Bay
AVAILABLE
4-Gate Pipeline Passed (Gate 1..4)\n[Escrow Locked, OTP Generated]
Reservation Cancelled\n[Escrow Refunded]
RESERVED_ESCROW
Bay Unlocked via 6-Digit OTP\n[Gear Picked Up by Renter]
Clean Inspection Approved\n[Rental Fee to Host, Collateral to Renter]
Damage Flagged & Slashed\n[Penalty Slashed to Host, Net Collateral to Renter]
ACTIVE_LEASE
Return Deposited in Locker Bay\n[Mandatory Quarantine Gate Locked]
INSPECTION_QUARANTINE
MANDATORY QUARANTINE INVARIANT:
Assets can NEVER transition from ACTIVE_LEASE directly to AVAILABLE.
Host/Inspector inspection is strictly required before collateral release.

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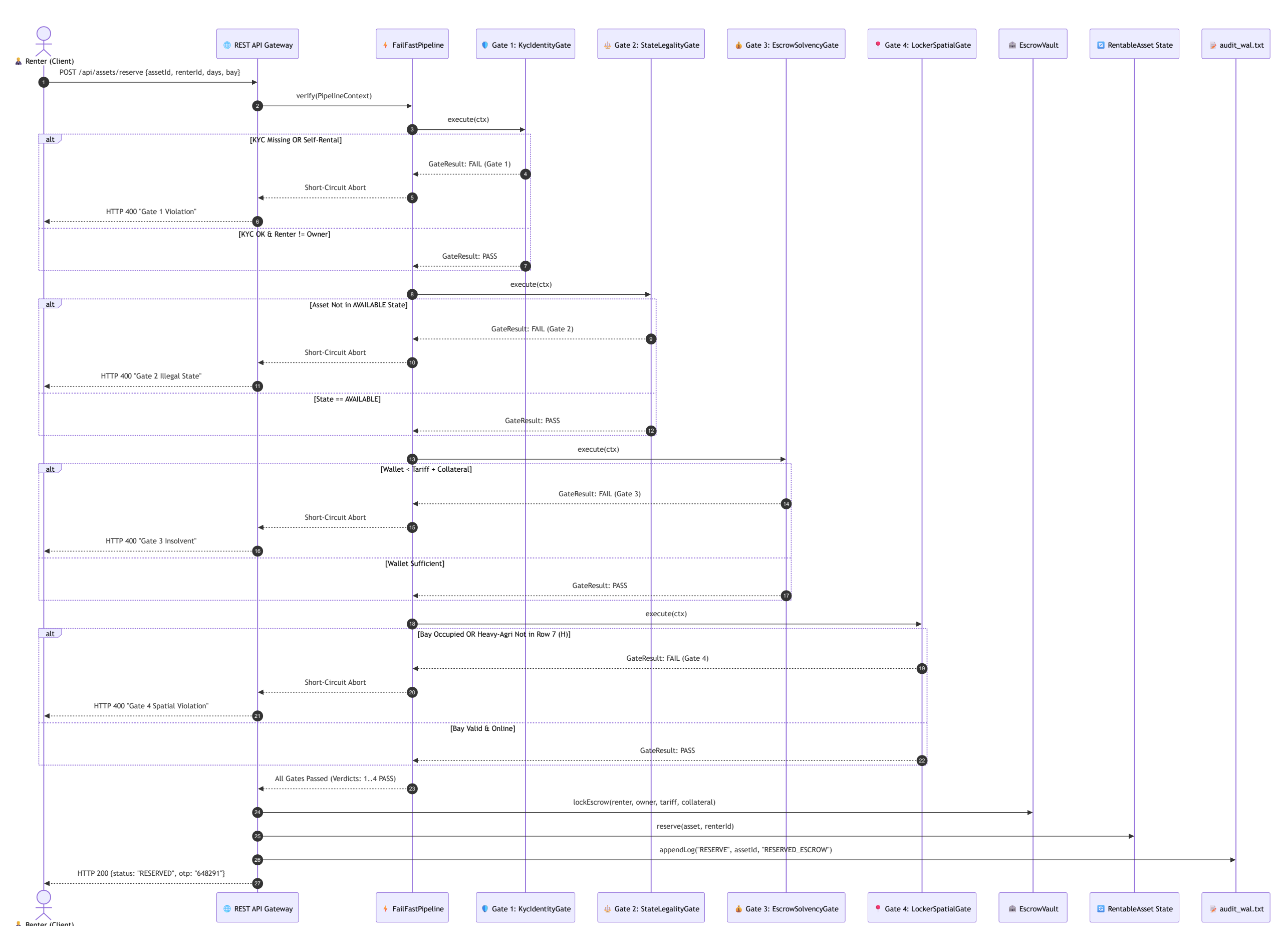
4. 4-Gate Fail-Fast Verification Pipeline Sequence Diagram

Demonstrates the short-circuiting sequential execution across `KycIdentityGate`, `StateLegalityGate`, `EscrowSolvencyGate`, and `LockerSpatialGate`.


Renter (Client)
REST API Gateway
FailFastPipeline
Gate 1: KycIdentityGate
Gate 2: StateLegalityGate
Gate 3: EscrowSolvencyGate
Gate 4: LockerSpatialGate
EscrowVault
RentableAsset State
audit_wal.txt
POST /api/assets/reserve {assetId, renterId, days, bay}
verify(PipelineContext)
execute(ctx)
alt
[KYC Missing OR Self-Rental]
GateResult: FAIL (Gate 1)
Short-Circuit Abort
HTTP 400 "Gate 1 Violation"
[KYC OK & Renter != Owner]
GateResult: PASS
execute(ctx)
alt
[Asset Not in AVAILABLE State]
GateResult: FAIL (Gate 2)
Short-Circuit Abort
HTTP 400 "Gate 2 Illegal State"
[State == AVAILABLE]
GateResult: PASS
execute(ctx)
alt
[Wallet < Tariff + Collateral]
GateResult: FAIL (Gate 3)
Short-Circuit Abort
HTTP 400 "Gate 3 Insolvent"
[Wallet Sufficient]
GateResult: PASS
execute(ctx)
alt
[Bay Occupied OR Heavy-Agri Not in Row 7 (H)]
GateResult: FAIL (Gate 4)
Short-Circuit Abort
HTTP 400 "Gate 4 Spatial Violation"
[Bay Valid & Online]
GateResult: PASS
All Gates Passed (Verdicts: 1..4 PASS)
lockEscrow(renter, owner, tariff, collateral)
reserve(asset, renterId)
appendLog("RESERVE", assetId, "RESERVED_ESCROW")
HTTP 200 {status: "RESERVED", otp: "648291"}

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5. 8x8 Smart Locker Bay Allocation Activity Diagram

Models physical bay sizing, coordinate computation, and Row 7 Heavy-Agri rule enforcement.

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Invalid Mermaid diagram
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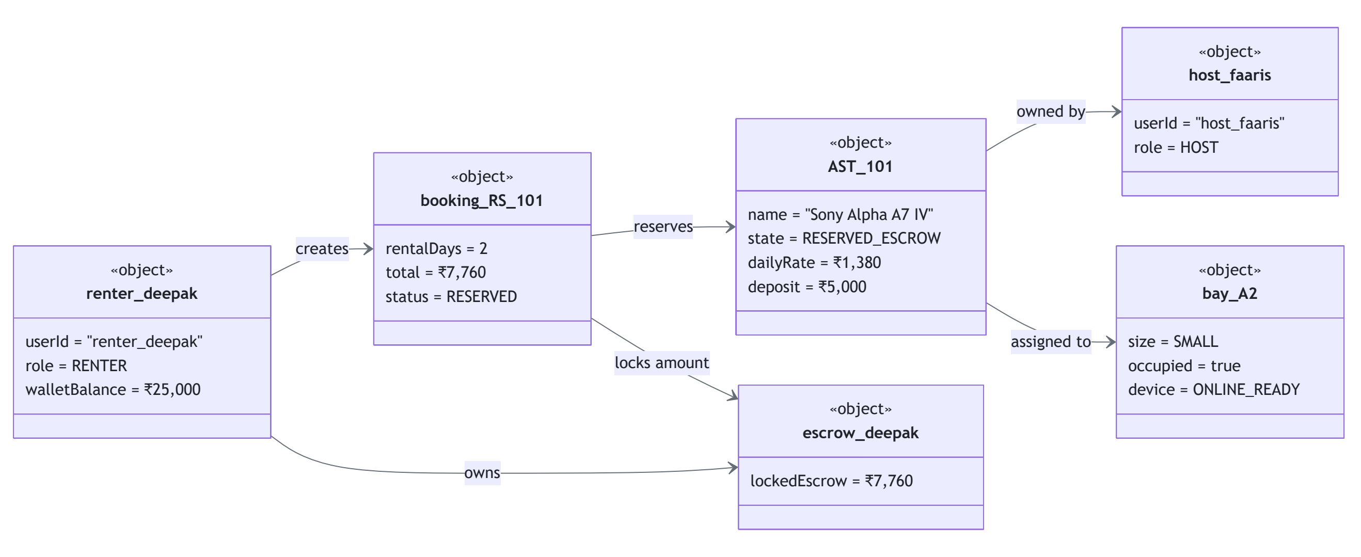
6. Object Diagram

This required UML diagram shows one realistic snapshot of objects during a rental. Unlike the class diagram, it shows actual sample instances and their links.



«object»
renter_deepak
userId = "renter_deepak"
role = RENTER
walletBalance = ₹25,000

«object»
booking_RS_101
rentalDays = 2
total = ₹7,760
status = RESERVED

«object»
AST_101
name = "Sony Alpha A7 IV"
state = RESERVED_ESCROW
dailyRate = ₹1,380
deposit = ₹5,000

«object»
host_faaris
userId = "host_faaris"
role = HOST

«object»
bay_A2
size = SMALL
occupied = true
device = ONLINE_READY

«object»
escrow_deepak
lockedEscrow = ₹7,760

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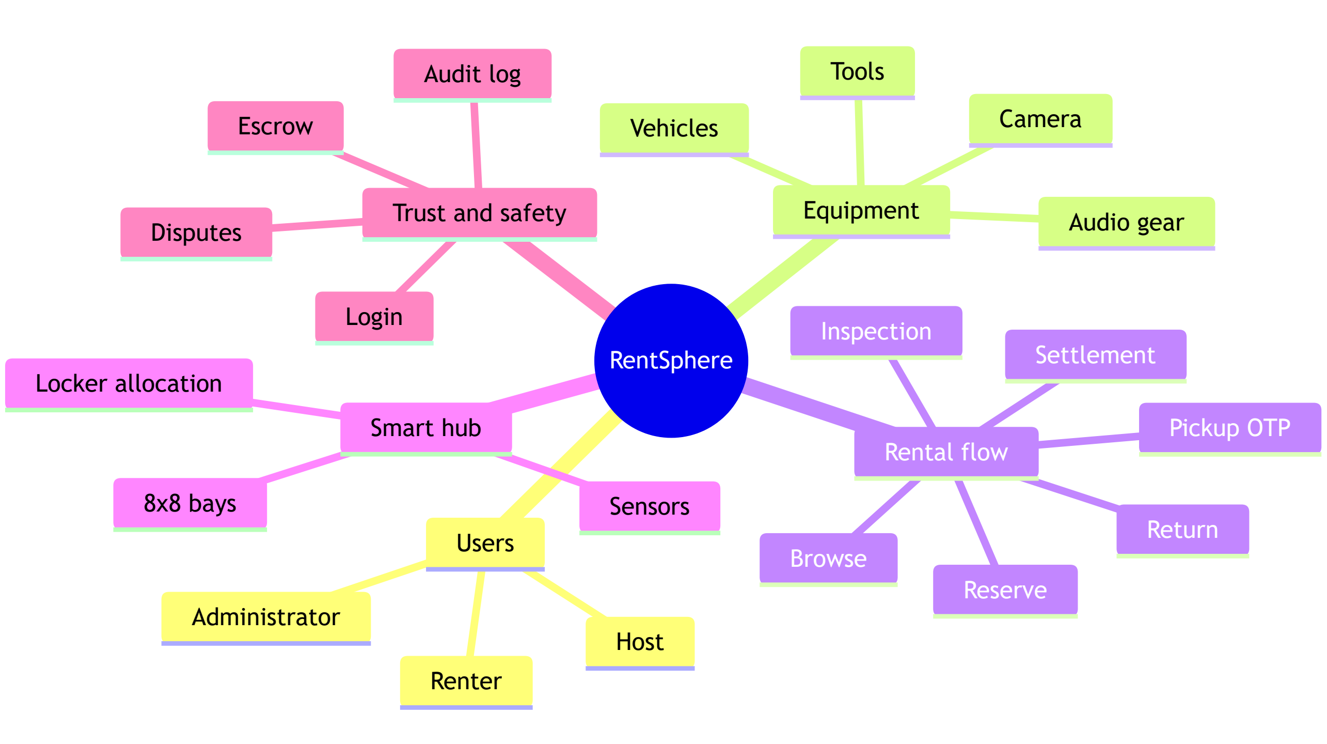
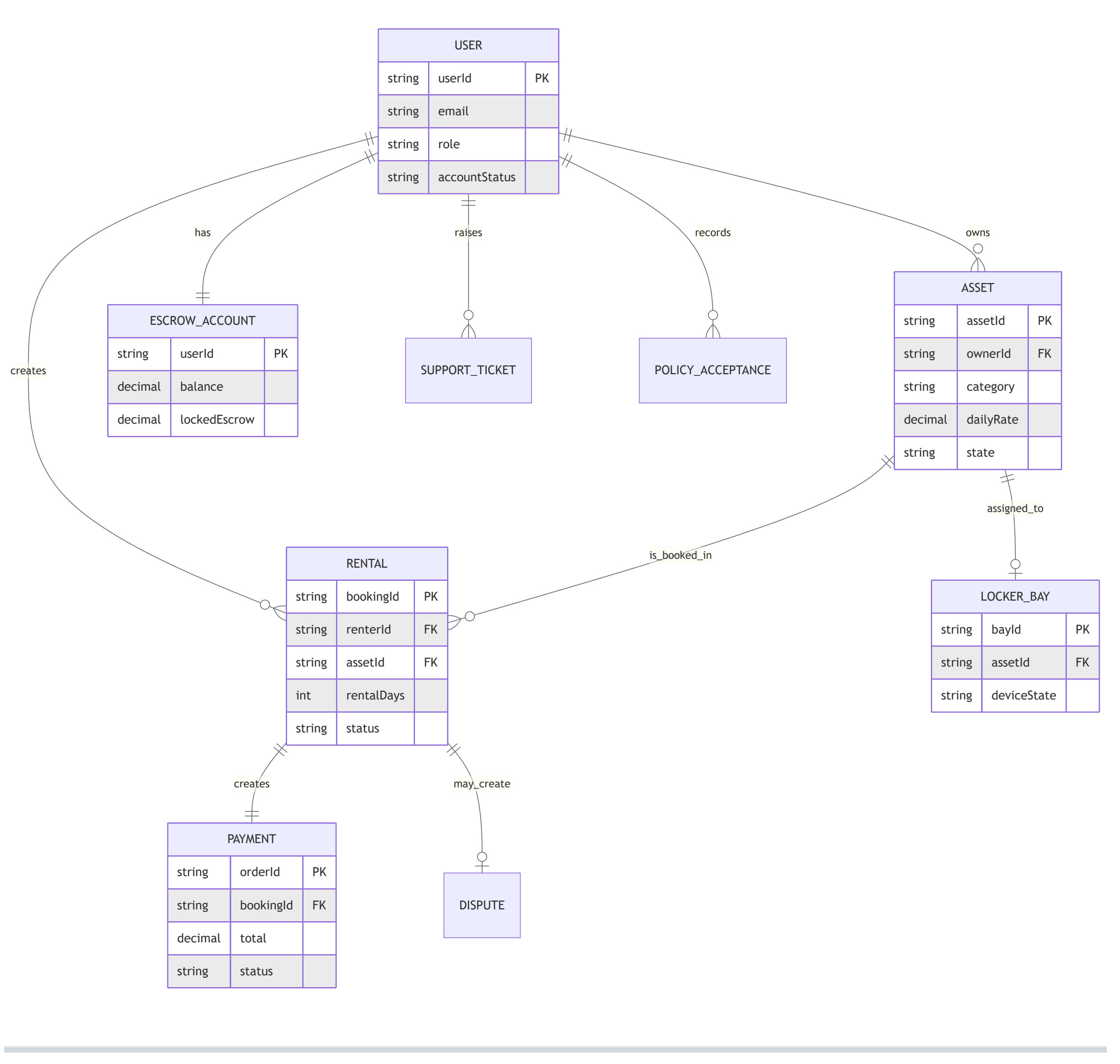
7. Brainstorming and Entity Relationship Diagram

7.1 Initial brainstorming map



7.2 Corresponding ER model



USER		
string	userId	PK
string	email	
string	role	
string	accountStatus	

ESCROW_ACCOUNT		
string	userId	PK
decimal	balance	
decimal	lockedEscrow	

ASSET		
string	assetId	PK
string	ownerId	FK
string	category	
decimal	dailyRate	
string	state	

RENTAL		
string	bookingId	PK
string	renterId	FK
string	assetId	FK
int	rentalDays	
string	status	

LOCKER_BAY		
string	bayId	PK
string	assetId	FK
string	deviceState	

PAYMENT		
string	orderId	PK
string	bookingId	FK
decimal	total	
string	status	

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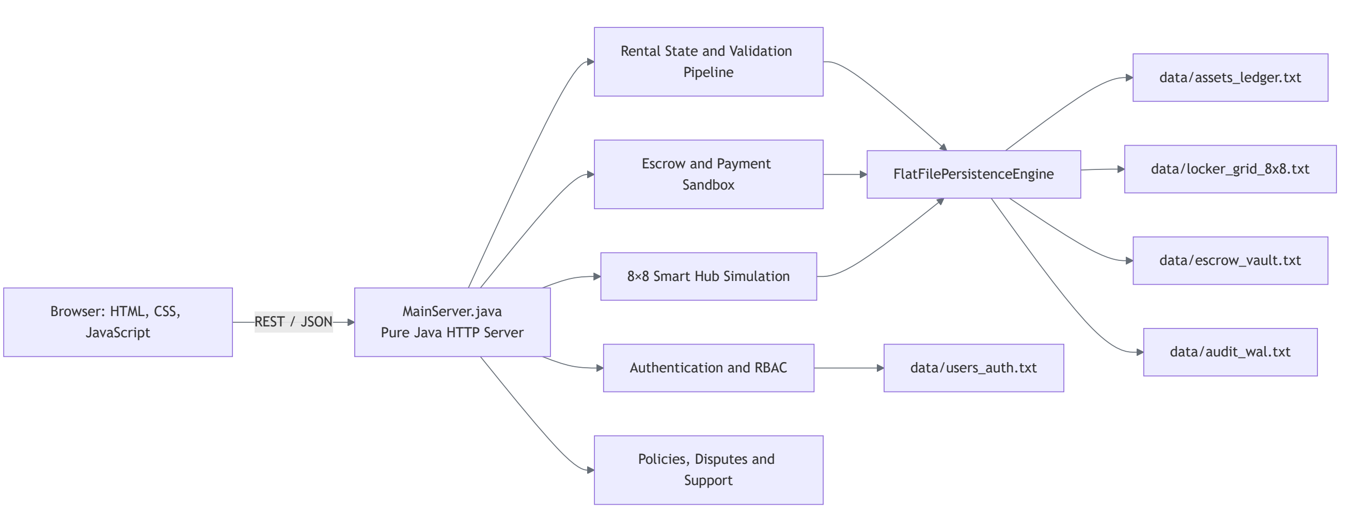
8. System Architecture Diagram



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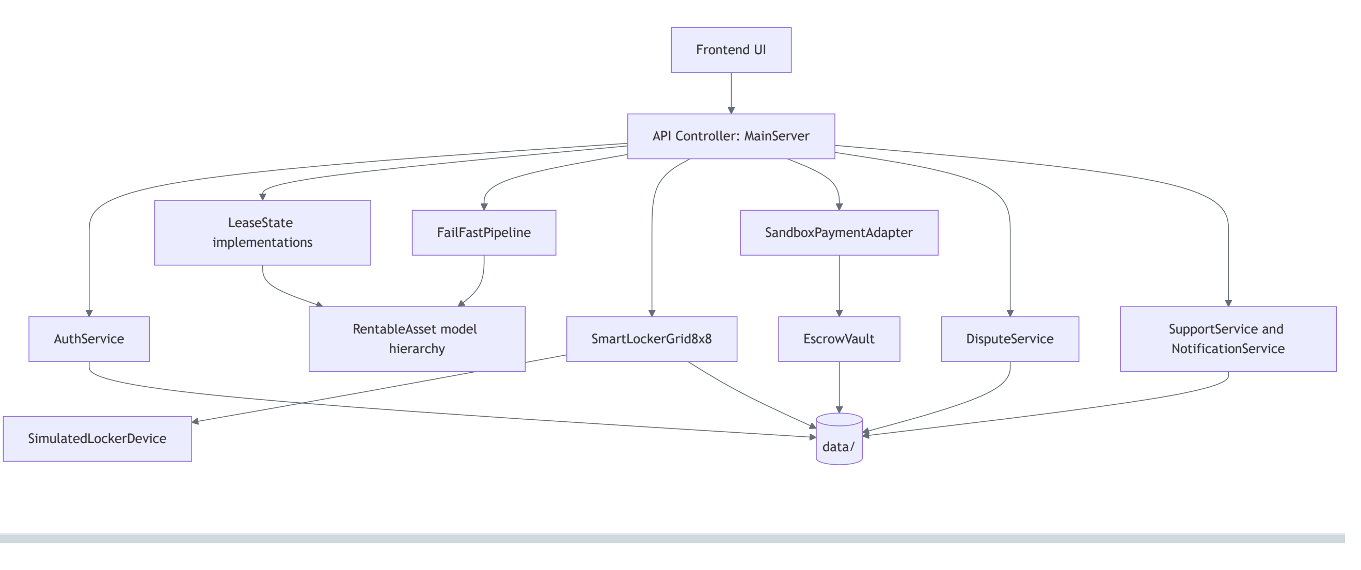
9. Component Diagram



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10. Local Deployment Diagram